Yimin Zhao

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PERSONAL SUMMARY

Interested in applying machine learning in autonomous systems. Strong project experience in neural network design, robot simulation, and robotic system development. Conducting PhD research of robotics at the National University of Singapore, supervised by Dr. Yeoh Ker-Wei, Justin.

Strength: Have strong execution and sense of responsibility, passionate to explore and learn.

EDUCATION

National University of Singapore, Singapore Feb. 2025 — Present
Doctor of Philosophy
Supervisor: Dr. Yeoh Ker-Wei, Justin
Interests: Machine Learning in Robotics; Neural Network; Autonomous Systems

National University of Singapore, Singapore Aug. 2023 — Feb. 2025
Master of Science in Robotics GPA: 4.35/5.0
Supervisor: Prof. Marcelo H. Ang Jr.
Representative Modules: Robot Vision and AI; Autonomous Mobile Robotics; Machine Learning in Robotics; Autonomous Robot Navigation; Robot Kinematics; Robot Dynamics and Control

University of Leeds, Chengdu, China Sept. 2019 — Jun. 2023
Bachelor of Science in Computer Science (2:1) Average Score: 72.2/100
Affiliation: SWJTU-Leeds Joint School, Southwest Jiaotong University
Representative Modules: Machine Learning; Algorithms and Data Structures; Artificial Intelligence; Data Mining; Object Oriented Programming; Software Engineering; Web Application Development

PUBLICATIONS

[1] Jack Wei Lun Shi, Wawan Solihin, Yufeng Weng, Houhao Liang, **Yimin Zhao**, Leong Hien Poh, and Justin K. W. Yeoh. Toward generalizable foundation models for 3D BIM geometry using a joint embedding predictive architecture. *Automation in Construction*, 191:107169, 2026.

[2] Jack Wei Lun Shi, Wawan Solihin, Yufeng Weng, **Yimin Zhao**, Leong Hien Poh, and Justin K. W. Yeoh. Self-supervised learning for BIM element classification using a joint embedding predictive architecture. *Automation in Construction*, 190:107075, 2026.

[3] Shuo Sun, **Yimin Zhao**, Christina Dao Wen Lee, Jiawei Sun, Chengran Yuan, Zefan Huang, Dongen Li, Justin K. W. Yeoh, Alok Prakash, Thomas W. Malone, and Marcelo H. Ang, Jr. AGI-Elo: How far are we from mastering a task? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, 2025.

[4] Chengran Yuan, Zijian Lu, Zhanqi Zhang, **Yimin Zhao**, Zefan Huang, Shuo Sun, Jiawei Sun, Jiahui Li, Christina Dao Wen Lee, Dongen Li, and Marcelo H. Ang, Jr. PIE: Perception and interaction enhanced end-to-end motion planning for autonomous driving. arXiv preprint arXiv:2509.18609, 2025.

[5] **Yimin Zhao** and Jin Gu. Feature fusion based on mutual-cross-attention mechanism for EEG emotion recognition. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024*, pages 276–285. Springer Nature Switzerland, 2024.

[6] Haohong Wang, Shenghao Gao, **Yimin Zhao**, Maojia Song, Heng Wang, and Damien Constantine Rompapas. The mind commands you: Combining brain-computer interactions with augmented reality to control internet of things (IoT) tools, and robotic platforms. In *2022 IEEE 5th International Conference on Electronics Technology (ICET)*, pages 1026–1031, May 2022.

ACADEMIC EXPERIENCE

Feature Fusion Based on Mutual-Cross-Attention Mechanism for EEG Emotion Recognition
National Natural Science Foundation of China, Singapore Dec. 2023 — Mar. 2024

- Proposed a purely mathematical MCA which fuse two features more effectively; developed a unique Channel-Frequency-Time 3D feature structure, which presents spectral and temporal information simultaneously.

- Published to Medical Image Computing and Computer Assisted Intervention – MICCAI 2024.

Motion Planning Simulation for Autonomous Driving
Master Thesis: Final Project Report, Singapore Oct. 2023 — Apr. 2024

- Simulated and visualized four advanced planners using the unified dataset and simulator provided by nuplan-devkit; created an evaluation score benchmark for comparison and analysis.
- Designed a novel learning-based planner based on diffusion model; simulated and evaluated the new planner for comparing with the benchmark.

Emotion Judgment System Based on Deep Learning and EEG Analysis
National Student Research Training Program, China May. 2021 — May. 2022

- Pre-processed DEAP dataset through band-pass filter and Independent Component Analysis (ICA) by using Python MNE package library.
- Extracted wavelet coefficients using the db4 wavelet of the continuous wavelet decomposition; generate 'scale' (64) dimension from frequency (128 Hz) dimension (Nyquist rate); calculated average energy to Shannon entropy ratio (EER) for each scale; selected appropriate ranges of scale to calculate.
- Built a novel four-classifier by merging bi-classifier; experimented and filtered the eight dominate channels with the most prominent emotional response to enhance model performance.

Design and fabrication of a brain-controlled rolling robot based on OpenBCI-Python-Arduino
Provincial Student Research Training Program, China Jun. 2020 — May. 2021

- Designed and programmed a simple 'admissible heuristic' by utilizing RNN in TensorFlow.
- Pre-processed EEG data using NeuroPype to remove noise; extracted feature through the Common Spatial Pattern (CSP); employed Linear Discriminant Analysis (LDA) as classifier.
- Designed and developed a novel 'Disk' human-computer interaction interface by PyQt of Python. The output of the classifier indirectly controlled the robot by switching 'gears' on the 'Disk' that represent different motion states.
- Published to International Conference on Electronics Technology (ICET), 2022.

WORK EXPERIENCES

Xi'an ZhenTec Co., Ltd
Software Engineer, Technical Department Jul. 2021 — Sept. 2021

- *EEG-based emotion classification:* Created video experimental paradigm; built a experimental platform utilizing marking box, EEG collector, amplifier, and paradigm; programmed data collection script.
- *EEG-based sleep stages monitor:* Implemented OSC port listening to achieve the overall data transmission; developed sleep stages display interface using PyQt.

EXTRACURRICULAR EXPERIENCES

School of Mechanical Engineering
President, Model Aircraft Association Sept. 2020 — Jun. 2022

- *Competition organization:* Planned and organised 12th and 13th Mechanics Innovation Competitions; responsible for the topics review, material and site preparation, publicity, and result review.
- *Vertical take-off and landing (VTOL) project:* Initiated the VTOL project sponsored by Leeds Life Foundation; designed structure and completed the deployment of the control devices.
- *Administrative work:* Took charge of the financial management, staff recruitment and documents writing.

HONOUR

- National Second Prize, China-US Young Maker Competition (CUYMC)
- Successful Participant, 2021 Mathematical Contest in Modeling

SKILLS

- **Programming:** Python, C++, MATLAB, C, HTML5, CSS3, JavaScript, Java
- **Libraries:** PyTorch, nuplan-devkit, TensorFlow, Numpy, Flask, Django, PyQt, Qt, OpenCV
- **Software:** LaTeX, Git, Linux, Anaconda, ROS Noetic, ROS Humble, Docker, OpenBCI, Eprime
- **Language:** Mandarin (native), English (proficient), Japanese (basic)